

Lab report 8

1. Objective

The objective of this lab is to understand and apply the concept of two-dimensional arrays in C++, including declaration, initialization, and accessing elements using nested loops.

2. Apparatus / Software

- Computer System
 - C++ Compiler (Code::Blocks / Dev C++ / VS Code)
-

3. Theory

A two-dimensional array is a collection of one-dimensional arrays arranged in rows and columns. It can be visualized as a table consisting of rows and columns.

Syntax:

```
type arrayName[x][y];
```

Here:

- `x` represents number of rows
- `y` represents number of columns

Each element is accessed using:

```
arrayName[i][j];
```

where `i` is row index and `j` is column index.

4. Lab Tasks Implementation

Task 1: Two-Dimensional Array Declaration, Initialization, and Display

```
#include <iostream>
using namespace std;

int main() {

    // 3x3 matrix declare aur initialize ki gayi hai
    int matrix[3][3] = {
        {1, 2, 3},
        {4, 5, 6},
        {7, 8, 9}
    };

    // nested loop ke zariye matrix ke tamam elements display karna
    for (int i = 0; i < 3; i++) {
        for (int j = 0; j < 3; j++) {
            cout << matrix[i][j] << " ";
        }
        cout << endl;
    }

    return 0;
}
```

Task 2: Sum of Rows, Columns, and Total Sum

```
#include <iostream>
using namespace std;

int main() {

    // 3x3 array random values ke sath initialize ki gayi hai
    int arr[3][3] = {
        {3, 6, 9},
        {2, 4, 6},
        {1, 5, 7}
    };

    int totalSum = 0;

    // har row ka sum calculate karna
    for (int i = 0; i < 3; i++) {
        int rowSum = 0;
        for (int j = 0; j < 3; j++) {
            rowSum += arr[i][j];
            totalSum += arr[i][j];
        }
    }
}
```

```

    }
    cout << "Row " << i + 1 << " sum: " << rowSum << endl;
}

// har column ka sum calculate karna
for (int j = 0; j < 3; j++) {
    int colSum = 0;
    for (int i = 0; i < 3; i++) {
        colSum += arr[i][j];
    }
    cout << "Column " << j + 1 << " sum: " << colSum << endl;
}

// total sum display karna
cout << "Total sum of all elements: " << totalSum << endl;

return 0;
}

```

Task 3: Finding Maximum Value in a 4x4 Array

```

#include <iostream>
using namespace std;

int main() {

    // 4x4 array random values ke sath initialize ki gayi hai
    int arr[4][4] = {
        {12, 5, 8, 19},
        {7, 25, 3, 14},
        {9, 11, 30, 6},
        {4, 18, 2, 22}
    };

    // pehle element ko maximum assume kiya gaya hai
    int maxValue = arr[0][0];

    // maximum value find karna nested loop ke zariye
    for (int i = 0; i < 4; i++) {
        for (int j = 0; j < 4; j++) {
            if (arr[i][j] > maxValue) {
                maxValue = arr[i][j];
            }
        }
    }

    // maximum value display karna
    cout << "Maximum value in array: " << maxValue << endl;

    return 0;
}

```

5. Conclusion

In this lab, two-dimensional arrays were implemented successfully. The tasks demonstrated declaration, initialization, traversal using nested loops, calculation of row and column sums, and finding the maximum element in a matrix.